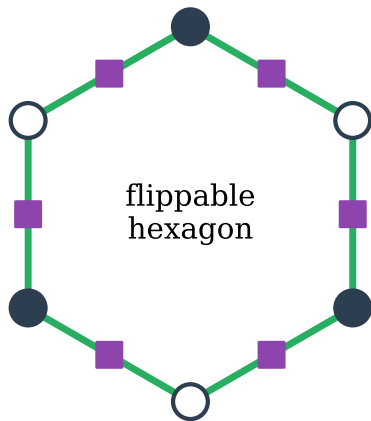
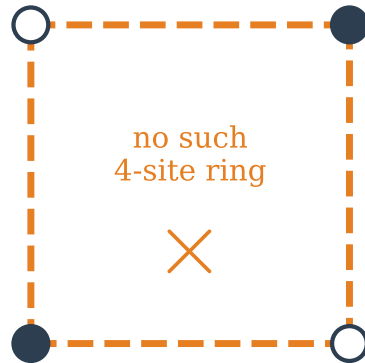


Why the smallest physical ring exchange has six spins

Allowed: 6-site ring



Forbidden in the bulk: 4-site ring



diamond lattice is bipartite $\Rightarrow$ shortest loop has 6 links
● tetrahedron (diamond vertex)

a 4-cycle needs two same-colour sites adjacent
(a diamond 4-loop) — it does not exist
■ spin (pyrochlore site = diamond link)